MIL-F-3040D

30 September 1976
SUPERSEDING
MIL-F-3040C
9 January 1967

# MILITARY SPECIFICATION

# FLAVORING TABLET, IMITATION VANILLA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for imitation vanilla flavoring tablets for use by the Armed Forces as an item of general issue.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| PPP-C-96 | - | Cans, Metal, 28 Gage and Lighter. |
| PPP-G-460 | - | Glass Containers, One Gallon Capacity and Smaller, For Other Than Medicinal Products, Packaging and Packing Of. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRXNM-TDS, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8950

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MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex

(Application for copies should be addressed to the Office of the Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

THE US PHARMACOPOEIAL CONVENTION, INC.

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, 20th and Northampton Sts., Easton, PA 18042.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

3. REQUIREMENTS

3.1 Materials. All of the materials shall be clean, sound and wholesome, and free from evidence of insect infestation or other foreign materials.



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3.1.1 Flavoring. Vanillin or ethyl vanillin may be used. Vanillin and ethyl vanillin shall be Food Chemicals Codex grade.

3.1.2 Lactose. Lactose shall be of edible grade.

3.1.3 Starch. Any starch of edible grade which will give the desired finished product may be used.

3.1.4 Binding and lubricating agents. Any edible binding and lubricating agent, normally used in tablet making and which will give a satisfactory finished product, may be used.

3.2 Finished product. The finished product shall comply with table I and 3.2.1.

TABLE I. Composition of tablets

| | Percentages | Weight (approx.) grains |
|---|---|---|
| Ethyl vanillin or | 10.3 ± 0.5 | 0.36 (23 mg) |
| Vanillin | 31.0 ± 1.5 | 1.08 (70 mg) |
| Starch -- not more than | 10.0 | 0.35 (0.23 mg) |
| Binding and lubricating agent, as needed | | |
| Lactose -- remainder | | |

* 3.2.1 Ingredients. The ingredients shall be pressed into tablets weighing 3.5 ± .1 grains (227 milligrams ± 6.5 mg). The moisture content shall be not more than 5.0 percent. The tablet shall disintegrate when examined in accordance with 4.2.2.2.

3.3 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder.

3.4 Workmanship. The product shall be prepared, processed and packaged under modern sanitary conditions in accordance with good commercial practice. The product shall be processed and packaged in establishments meeting the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor



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may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Preacceptance inspection (plant sanitation inspection). Award of contracts shall be limited to plants which have received prior sanitary approval by the Military Veterinary Services in accordance with MIL-STD-668.

4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Examination of ingredients. The sample unit shall be approximately eight ounces (224 g) of product taken from a primary container. The sample size shall be five sample units. In the event a lot consists of less than five containers, a sample unit shall be drawn from each container in the lot. Each ingredient shall be examined organoleptically as necessary, to determine compliance with condition requirements of 3.1. In addition compliance of vanillin or ethyl vanillin to grade requirements of 3.1.1 and to identity requirements of all other listed materials under 3.1, shall be ascertained by examining pertinent labels, invoices or other valid documents. Noncompliance shall indicate an unacceptable ingredient, and use of such shall be cause for rejection of the involved finished product.

4.2.1.2 Testing of components. Testing of components described in this specification shall be as shown in table II. Results shall apply to the lot average.

TABLE II. Testing of components

| Component | Sample unit | Lot size expressed in | Inspection level | Characteristics | Results reported | Test ref & rqmt para |
|---|---|---|---|---|---|---|
| Metal cap | Six caps | Caps | S-1 | Tin plate | Nearest 0.01 lb/ base box | 4.3.3 5.1 |



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* 4.2.2 Examination of end item. Examination of the end item shall be performed in accordance with tables IV through VIII amd 4.2.2.1. The examination criteria for the aforementioned tables shall be as shown in table III. AQLs for tables IV and VII shall be expressed in terms of defects per hundred units and the AQLs for tables V, VI and VIII shall be expressed as percent defective.

TABLE III. End item examination criteria

| Table | Inspection level | Sample unit | Lot size expressed in | AQLs | | | |
|---|---|---|---|---|---|---|---|
| | | | | Major A | Major B | Minor | Total |
| IV | 1 | One primary container | Primary containers | 0.25 | 1.5 | — | 4.0 |
| V | S-3 | One tablet from a primary container | Primary containers | — | — | 4.0 | — |
| VI | S-2 | Contents of a primary container | Primary containers | — | — | 4.0 | — |
| VII | S-2 | Ten tablets from a primary container | Primary containers | — | 4.0 | — | 10.0 |
| VIII | S-2 | One filled and sealed primary container | Primary containers | — | 2.5 | — | — |

TABLE IV. Examination of primary container

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| 101 | | | Broken or cracked. |
| 102 | | | Not closed. |



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TABLE IV. Examination of primary container (cont'd)

| Category | | | Defect |
|---|---|---|---|
| Major A | Major B | Minor | |
| | 151 | | Closure cap not tight. 1/ |
| | 152 | | Not style or size bottle specified. |
| | 153 | | Chipped or thin spot. |
| | 154 | | Cotton or rayon plug missing or insufficient to prevent movement of tablets. |
| | 155 | | Cotton or rayon has odor not associated with product. |
| | 156 | | Pitted rust on cap. 2/ |
| | 157 | | Interior or exterior enamel coating of cap missing, not uniform or not corrosion-resistant. 3/ |
| | 158 | | Nomenclature missing, incorrect or illegible. |
| | 159 | | Directions for use or strength of tablets information missing, incorrect or illegible. |
| | | 201 | Not clean. |
| | 160 | | Cap or cap liner missing, improperly positioned, damaged or unsuitable for the product. |
| | | 202 | Cap style not as specified. |
| | | 203 | Labeling information other than nomenclature missing, incorrect or illegible. |
| | | 204 | Label not hardsize. |

1/ For determining tightness of closure for screw-type cap, the following is applicable: Place the primary container upright on top of a filled and closed fiberboard case. Grasp the side of the cap firmly with the fingertops and turn cap gently counter-clockwise with a minimum of downward pressure for one complete revolution. If the cap turns and the container remains stationary, a defect shall be scored.

2/ Rust that can be removed by wiping with a soft cloth will not be considered as pitted rust.

3/ Corrosion-resistant enamel shall be ascertained by examining pertinent labels, invoices or other valid documents. Noncompliance with this requirement shall be cause for rejection of the lot.



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TABLE V. Examination for net weight of tablets 1/

| Category | Defect |
|---|---|
| Minor | |
| 201 | Tablet less than 3.4 or more than 3.6 grains (220-227 milligrams). |

1/ Results shall be reported to the nearest 0.1 grain.

TABLE VI. Examination for count in primary containers 1/

| Category | Defect |
|---|---|
| Minor | |
| 201 | Count is less than 186 tablets. |

1/ Lot shall be rejected if sample data indicates lot average count is less than specified net count.

TABLE VII. Examination of product characteristics 1/

| Category | | Defect |
|---|---|---|
| Major B | Minor | |
| 151 | | Tablet broken or split. |
| | 201 | Tablet has uneven face(s) or edge(s). |

1/ Presence of foreign material (e.g., rust, dirt, grease, weed particles, paint, enamel, insects, insect fragments, glass, hair, metal, or other foreign materials), shall be cause for rejection of the entire lot.



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TABLE VIII. Examination for water resistance of labels 1/

| Category | Defect |
|---|---|
| Major B | |
| 151 | Label on primary containers displaced with finger pressure. |

1/ Water resistant test shall be conducted in accordance with 4.3.2.

4.2.2.1 Examination of packaging, packing and marking. Examination of packaging, packing and marking shall be in accordance with the examination criteria of PPP-G-460.

* 4.2.2.2 Examination of tablet. The tablet when dropped into a fluid ounce (30 ml) of water at a temperature of 75°F (24°C) shall disintegrate with a gentle manual agitation in approximately one minute.

4.2.3 Sampling procedure and acceptance criteria for testing of finished product. Procedures for testing for moisture, vanillin or ethyl vanillin, starch, and disintegration, shall be in accordance with 3.2, 3.2.1 and 4.3.1. Requirements shall be applicable to the lot average, except the requirement for disintegration, which shall be a unit requirement. The sample for testing for moisture, vanillin or ethyl vanillin and starch, shall be a composite of 400 tablets derived from the number of primary containers indicated by inspection level S-3. The sample unit for testing for disintegration shall be one tablet from a bottle, and the inspection level shall be S-3 with an acceptance quality level of 4.0, expressed as percent defective. Lot size shall be expressed in terms of primary containers. Any unit requirement test result not within specified limits shall be considered a defect. Composite test results shall be reported to the nearest 0.1 percent. Noncompliance to one or more test requirements shall be cause for rejection of the lot.

4.3 Tests. Test procedures and controls which differ from those specified herein may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.3.1 Chemical analyses. Chemical analyses shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists:



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| | | |
|---|---|---|
| Vanillin or ethyl vanillin | Chapter: Flavors<br>Section: Vanilla Extract and its Substitutes | Coumarin Official Photometric Method or Vanillin, Ethyl Vanillin and Coumarin, Official Chromotographic Separation Method. |
| Starch | Chapter: Animal Feed<br>Section: Starch | Direct Acid Hydrolysis |
| Moisture | Chapter: Animal Feed<br>Section: Moisture | Distillation with Toluene (use of 50 gr. sample) |

4.3.2 Water-resistance test for label adhesive. The label adhesive shall be tested for water resistance as follows: Submerge glass container with label affixed in room temperature water for four hours. Remove from water and blot with paper towel. Press finger firmly against label (in an area where adhesive has been applied) and move finger (firmly pressed against the label) towards the bottom of the bottle. Displacement of the label indicates failure to meet the requirement.

4.3.3 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PACKAGING

5.1 Preservation-packaging, level A. One hundred and ninety-two tablets shall be packaged into a round or rectangular, 2-ounce (59 ml), widemouth bottle with a continuous thread finish. The cap closure shall be a curled edge, knurled cap made from not less than commercial 0.25-pound per base box electrolytic tin plate (see 4.3.3) and shall be coated inside and outside with a suitable corrosion-resistant enamel. The cap shall be fitted with a liner suitable for the product. The cap shall be tightly affixed in position. A sufficient quantity of odorless cotton or rayon shall be placed on top of the tablets to prevent movement of the tablets.

5.1.1 Intermediate packaging. Twelve bottles shall be packaged into a paperboard box in accordance with PPP-G-460.

5.2 Packing, levels A and B. Packing shall be in accordance with the level A or B requirements, as specified (see 6.1) of PPP-G-460, except that only four intermediate packages shall be packed into a shipping container and that the shipping container shall be reinforced with only one strap, applied girthwise.



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5.3 Labeling and marking.

5.3.1 Glass containers. The required information shall be printed on a hard-sized paper label. The label shall be fixed in position with a water-resistant adhesive (see 4.3.2). The printed labels shall contain the following information:

192 IMITATION VANILLA FLAVORING TABLETS

(In letters larger than any other used for the label)

Equivalent to one quart of regular extract.
One tablet is equal to one teaspoonful or 1/6 ounce of regular extract.
Three tablets equal one tablespoonful or 1/2 ounce of regular extract.

Directions for use

Dissolve the required number of tablets in a portion of any liquid used in the recipe.
Any other information required by the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.
Name and address of manufacturer (in type smaller than any of the above).

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and PPP-G-460.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing required (see 5.2).

6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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6.3 Award of contracts for the product specified in this document shall be limited to plants known to maintain the required sanitation conditions of 3.4 (also see 4.1.1).

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD, GL
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

Project No. 8950-0151



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MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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